

The two-person and zero-sum matrix game with probabilistic linguistic information

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ARTICLE INFO

Article history:

Received 31 October 2020

Received in revised form 19 March 2021

Accepted 10 May 2021

Available online 13 May 2021

Keywords:

Matrix game

Multiple criteria analysis

Probabilistic linguistic information

Membership function

Two-person

Zero-sum

ABSTRACT

Game theory establishes mathematical models of strategic interactions among rational decision-makers. In many situations, the crisp values of payoffs are difficult to obtain while the probabilistic linguistic information is easy to collect. However, the existing research on matrix game did not consider the probabilistic linguistic information. To bridge this gap, this study takes the probabilistic linguistic information as the input of a two-person and zero-sum matrix game, and addresses the vague information by triangular membership functions. Such a two-person and zero-sum matrix game with probabilistic linguistic information is a useful technique for multiple criteria analysis. In addition, it outputs the same form of information as the inputs to increase the interpretability compared with other uncertain matrix games. An illustrative example about the forest management is provided to show the validity and advantages of the two-person and zero-sum matrix game with probabilistic linguistic information.

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1. Introduction

Game theory studies the strategies of rational individuals under specific conditions [27]. In a game, players are required to provide payoff values for different strategies in order to formulate a game model. The number of payoff values that need to be determined increases significantly with the increase of the numbers of players and strategies. Consequently, requiring players to provide accurate payoff values would bring a heavy burden to players while the uncertain information or rough estimation is a feasible and flexible alternative for payoff values in the game. This study focuses on the two-person matrix game with uncertain information, which only considers two players and the payoff matrix is symmetric. The investigation can be characterized by three aspects: the technique for modelling uncertain information, the technique for defuzzifying uncertain information, and the applications of two-person matrix games.

Intervals, fuzzy numbers and linguistic models are three main categories for modelling uncertain information in a two-person matrix game. The first type of uncertain information has been studied in [18,22] from the perspective of intervals' endpoints. The fuzzy information has been modelled by triangular fuzzy numbers [2,6,7,9,17,23,34], trapezoidal fuzzy numbers [16,19,20], intuitionistic fuzzy numbers [13,26], hesitant fuzzy numbers [36] and dual hesitant fuzzy numbers [15]. Single linguistic terms [1], 2-tuple linguistic terms [33] and interval 2-tuple linguistic terms [32] are widely-used linguistic models. The probabilistic linguistic information is a powerful and flexible information form which aligns human thoughts

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[21,28,35], but, to the best of our knowledge, the probabilistic linguistic information has not been used in the existing matrix game models. This is the first research gap intended to be filled by this study in terms of the modelling uncertain information in matrix games.

The uncertain information in a matrix game can be tackled by defuzzification techniques. According to the categories of uncertain information, we can summarize the existing researches from three perspectives. Interval values were introduced to the uncertain game by their endpoints [18,22], and the obtained game goals were intervals as well. The fuzzy game defined in [6] was studied in terms of either the membership function [2,7,17,19,20,23,34,36] or the membership and non-membership functions together [3,13,15,26]. Triangular fuzzy numbers [2,7,17,23,34] and trapezoidal fuzzy numbers [16,19,20] were two commonly-adopted membership functions. To defuzzify the above fuzzy information, the ranking functions of fuzzy numbers [2,7,34], extension principle [23], alpha cut interval [17,19,20], average indices of membership degrees and non-membership degrees [26], logarithmic values of fuzzy numbers' product [8], value function and ambiguity function [36], and similarity index [15] were used in different scenarios. The values of a game with fuzzy information could be converted to crisp goals [2,7,15,36] and fuzzy goals [13,17,19,20,23,26,34], respectively. [1,5,32,33] fused linguistic terms in a game without introducing membership functions, which may loss the information of uncertainty. Membership functions are fundamental in defuzzifying the probabilistic linguistic information in a matrix game. However, as far as we know, there is little research regarding the membership functions of probabilistic linguistic information. This is the second research gap of this study.

The two-person matrix game is not only a useful tool to analyze the strategies of two players, but also an advantageous technique to implement the game between the player and Nature [31]. Chen and Larbani [8] used the game with Nature to solve the selection of optimal titanium dioxide product development for an enterprise. Singh and Gupta [33] adopted the 2-tuple linguistic information in the game with Nature to select the optimal air conditioning system for installation. In these examples, players were advised to adopt the pure strategy. In other words, only one strategy can be employed in these games, rather than the mixed strategies in the matrix game. This is the third research gap that this study intends to solve.

Motivated by the above analysis, this study introduces a two-person and zero-sum matrix game with probabilistic linguistic information, which increases the feasibility, flexibility and application scope of the game theory and reduces the burden of players evolved in the game. In addition, the triangular membership functions of probabilistic linguistic information are presented to defuzzify linguistic evaluations. The retranslation process from membership degrees to the corresponding probabilistic linguistic information is proposed to enable the interpretability of the output results. Furthermore, the probabilistic linguistic two-person and zero-sum matrix game is adopted to solve a multiple criteria decision-making problem from the perspectives of a player and Nature, which provides a feasible way for multiple criteria analysis without weight information.

The rest of this paper is organized as follows: Preliminaries of the two-person and zero-sum game are provided in Section 2. Section 3 presents the solutions to the two-person and zero-sum matrix game with probabilistic linguistic information. An example about the forest management is given in Section 4. Section 5 ends this paper with concluding remarks.

2. Related works of two-person and zero-sum game

In this section, the main characteristics of a two-person and zero-sum game are introduced first and the literature review of related works is given.

2.1. Description of two-person and zero-sum game

Competition exists in most realistic games. The zero-sum game is a fundamental game with two players. A two-person and zero-sum game G can be considered as a triplet $G(\mathbf{S}^m, \mathbf{S}^n, \mathbf{V})$, where $\mathbf{S}^m$ and $\mathbf{S}^n$ are the strategy space of the player I and the player II, respectively, and $\mathbf{V}$ is an $m \times n$ matrix with payoff values of the player I. In a two-person and zero-sum game, the payoff matrix of the player II is $-\mathbf{V}$, which ensures the total game value being zero. The player I and the player II are designed to maximize benefits and minimize costs, respectively.

Let x and y be the strategies of the player I and the player II, which belongs to the non-negative orthant of the m -dimensional and n -dimensional Euclidean space, respectively. As for the strategy spaces $\mathbf{S}^m$ and $\mathbf{S}^n$, a special vector $\mathbf{e}^T$ with each component being one was introduced. That is, $\mathbf{e}^T = (1, \dots, 1)$ and the dimension of $\mathbf{e}^T$ could be $m \times 1$ and $n \times 1$ for different strategy spaces. Then, the strategy spaces $\mathbf{S}^m$ and $\mathbf{S}^n$ can be represented as $\mathbf{S}^m = \{x \in \mathbf{R}^{m+}, \mathbf{e}^T x = 1\}$ and $\mathbf{S}^n = \{y \in \mathbf{R}^{n+}, \mathbf{e}^T y = 1\}$, respectively. The payoff values of the player I can be obtained by the scalar $x^T \mathbf{V} y$.

The solution of a two-person and zero-sum game is defined as $G(x^*, y^*, v^*)$. If for any $y \in \mathbf{R}^{n+}$, $(x^*)^T \mathbf{V} y \geq v^*$ and for any $x \in \mathbf{R}^{m+}$, $x^T \mathbf{V} y^* \leq v^*$, then, v^* is the optimal value of the game G , and x^* and y^* is the optimal strategy for the player I and the player II, respectively. (x^*, y^*) is also referred as the saddle point of the game G .

2.2. Literature review on the two-person and zero-sum game

In classic game theory, it is commonly assumed that each player's payoff or utility function is accurately known and its values are crisp values. However, considering the complexity of conflicts, it is often difficult to provide crisp payoff values in many applications, and, as a result, the two-person game with uncertain payoff values has attracted lots of attention. The fuzzy game was first proposed by Butnariu [6] to denote each player's beliefs over other players' strategies. Campos [7] adopted a ranking function of fuzzy numbers to defuzzify the fuzzy input information, which requires the prior knowledge of players in the game. Bector, Chandra and Vijay [2] used another ranking function of fuzzy numbers to defuzzify fuzzy inequalities and employed the crisp goals of two players. Furthermore, Vijay, Chandra and Bector [34] modified crisp goals into fuzzy goals and used the similar defuzzification function to solve a fuzzy game. Lui and Kao [23] solved a fuzzy game with eleven different alpha-cut intervals of triangular fuzzy payoff values. Similarly, Li [17] adopted three points obtained by the alpha values being zero to establish multi-objective models of two players and solved them by a lexicographic method. Afterwards, Li and Xuan [20] used a similar method as that of Li [17] to solve a matrix game with trapezoidal fuzzy numbers. Li [18] and Liu, and Tao [22] used interval endpoint values to establish four objectives for two players in a non-cooperative game. Li [19] generalized the alpha value into the continuous interval $[0, 1]$ instead of previous separate values, zero and one. Xue, Xu and Zeng [36] used the ambika method to obtain the crisp goal value and ambiguity goal value of a game. Besides the membership function, the non-membership function was also introduced in the forms of intuitionistic fuzzy information [13,26] and dual hesitant fuzzy information [15]. The intuitionistic fuzzy information was respectively transformed into the average indices of the membership and non-membership degrees [296], and the logarithm values of the products of intuitionistic fuzzy values [13] in the goal of a game. The dual hesitant fuzzy numbers were converted by a similarity measure in a game [9]. Bhaumik, Roy and Li [4] used the alpha cut to capture the neutrosophic information in a bi-matrix game. Roy and Maiti [30] adopted type-2 fuzzy variables in a Stackelberg game and solved it by the genetic algorithm. The linguistic information was incorporated in the game with single linguistic terms [1]. Later, the 2-tuple linguistic information [33] and interval 2-tuple linguistic information [22] were integrated in the game by considering the mean values of the membership functions of linguistic terms.

From the aspect of information representation form, the probabilistic linguistic information, a flexible tool for computing with words, has not been investigated within the framework of game theory. For a matrix game with linguistic information, the symbolic computation instead of the operations on membership functions may lose the semantics of linguistic terms in the computation process. In addition, in some existing fuzzy games [2,7,15,36], the goals of these fuzzy games were supposed to be crisp, which were not consistent with the uncertainty of the game. In this regard, Section 3.2 will present a membership function for the probabilistic linguistic information to depict and retranslate the probabilistic linguistic information of a game.

Regarding the applications of two-person matrix games, two multiple criteria decision examples were implemented by the game between the player and Nature with fuzzy information [31] and 2-tuple linguistic information [33], respectively. The applicability of the two-person matrix game between the player and Nature in the multiple criteria analysis with probabilistic linguistic information is the third motivation of this study. Section 4 provides an example about forest management by considering Nature as the player II.

3. The two-person and zero-sum matrix game with probabilistic linguistic information

This section introduces the two-person and zero-sum matrix game with probabilistic linguistic information. We first introduce the terminology of a probabilistic linguistic matrix game in Section 3.1. Then, the triangular fuzzy membership functions are presented to defuzzify the probabilistic linguistic information. After that, probabilistic linguistic linear programming is proposed to find the optimal strategies of players and the value of the probabilistic linguistic game. The procedure of our method can be visualized in Fig. 1.

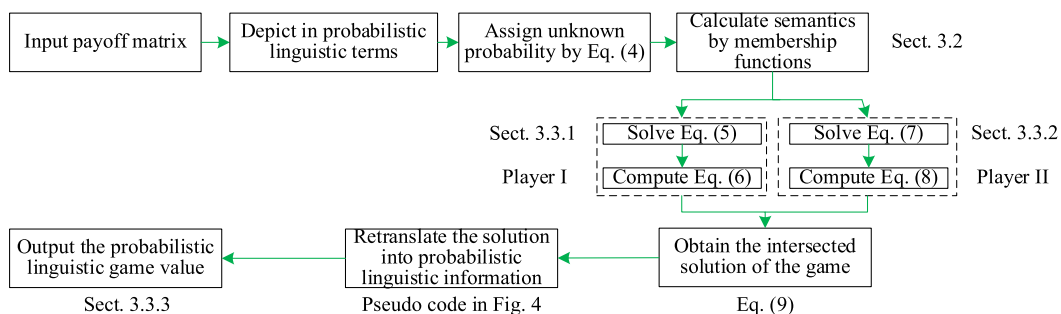


Fig. 1. The procedure of the two-person and zero-sum matrix game with probabilistic linguistic information.

3.1. Terminology of a probabilistic linguistic matrix game

In uncertain situations, the probabilistic linguistic information can be introduced to model uncertain payoff values. The payoff matrix $\tilde{\mathbf{V}}$ with probabilistic linguistic information can be denoted as:

$$\tilde{\mathbf{V}} = \begin{pmatrix} \tilde{v}_{11} & \tilde{v}_{12} & \cdots & \tilde{v}_{1m} \\ \tilde{v}_{21} & \tilde{v}_{22} & \cdots & \tilde{v}_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ \tilde{v}_{n1} & \tilde{v}_{n2} & \cdots & \tilde{v}_{nm} \end{pmatrix} \quad (1)$$

where $\tilde{v}_{ij}$ is a probabilistic linguistic term set, for $i = 1, 2, \dots, m, j = 1, 2, \dots, n$.

Considering the ambiguity in $\tilde{\mathbf{V}}$, the constraints and final values of the probabilistic linguistic game would be uncertain. Further the player I and the player II are also considered from the aspects of the gaining floor and the lost ceiling, respectively. The goal of each player can be established respectively as follows:

$$\text{Player I : } \max_{\mathbf{x} \in S^m} \min_{\mathbf{y} \in S^n} \mathbf{x}^T \tilde{\mathbf{V}} \mathbf{y}; \text{ Player II : } \min_{\mathbf{y} \in S^n} \max_{\mathbf{x} \in S^m} \mathbf{x}^T \tilde{\mathbf{V}} \mathbf{y}$$

Let $\tilde{v}_1$ denote the minimal fuzzy payoff value of the player I. $\mathbf{x}^T \tilde{\mathbf{V}} \mathbf{y}$ can be viewed as a weighted average of the expected payoff value of the player I if (s)he uses $\mathbf{y} \in S_n$ against player II's pure strategy. The pure strategy of the player II can be

denoted as $\mathbf{y} = \left(\overbrace{0, \dots, 0}^{j-1}, \overbrace{1, 0, \dots, 0}^{n-j} \right)^T$. The goal of the player I can be converted as $\mathbf{x}^T \tilde{\mathbf{V}} \mathbf{y} \geq \tilde{v}_1$, for every strategy $\mathbf{y} \in S_n$. Sim-

ilarly, let $\tilde{v}_2$ represent the maximal fuzzy payoff value of the player II. $\mathbf{x}^T \tilde{\mathbf{V}} \mathbf{y}$ can be regarded as a weighted average of the expected payoff value of the player II if (s)he uses $\mathbf{x} \in S_m$ against player I's pure strategy $\mathbf{x} = \left(\overbrace{0, \dots, 0}^{i-1}, \overbrace{1, 0, \dots, 0}^{m-i} \right)^T$. Then,

the goal of the player II is: $\mathbf{V} \mathbf{y} \leq \tilde{v}_2$, for each strategy $\mathbf{x} \in S_m$. For each element in the matrix $\tilde{\mathbf{V}}$, we have for the player I

$$\begin{aligned} & \max \{ \tilde{v}_1 \} \\ \text{s.t. : } & \begin{cases} \sum_{i=1}^m \tilde{v}_{ij} x_i \geq \tilde{v}_1, j = 1, 2, \dots, n \\ \sum_{i=1}^m x_i = 1; x_i \geq 0, i = 1, 2, \dots, m \end{cases} \end{aligned} \quad (2)$$

where the fuzzy objective value $\tilde{v}_1$ can be modelled as $(a_{\tilde{v}_1} - b_{\tilde{v}_1}, a_{\tilde{v}_1}, a_{\tilde{v}_1} + c_{\tilde{v}_1})$, and for the player II,

$$\begin{aligned} & \min \{ \tilde{v}_2 \} \\ \text{s.t. : } & \begin{cases} \sum_{j=1}^n \tilde{v}_{ij} y_j \leq \tilde{v}_2, i = 1, 2, \dots, m \\ \sum_{j=1}^n y_j = 1; y_j \geq 0, j = 1, 2, \dots, n \end{cases} \end{aligned} \quad (3)$$

where the fuzzy objective value $\tilde{v}_2$ can be modelled as $(a_{\tilde{v}_2} - b_{\tilde{v}_2}, a_{\tilde{v}_2}, a_{\tilde{v}_2} + c_{\tilde{v}_2})$.

3.2. Membership functions of probabilistic linguistic term sets

After establishing the mathematical models of a probabilistic linguistic matrix game, we then address the membership function of each linguistic term in this subsection so as to defuzzify the probabilistic linguistic information. To take the advantages of linearity, simplicity, interpretability, and consistency, the triangular fuzzy membership functions of linguistic terms are used. When describing the membership functions of the probabilistic linguistic information $\tilde{\mathbf{V}}$, the universe of discourse U and the granularity of the linguistic term set are required as pre-known information. For example, if the granularity of a linguistic term set is nine and the universe of discourse U is from zero to ten, then the triangular membership functions of nine linguistic terms, $s_0, s_1, \dots, s_7, s_8$, are shown in Fig. 2. The granularity Γ of this linguistic term set $\{s_0, s_1, \dots, s_7, s_8\}$ is nine as there are nine linguistic terms from s_0 to s_8 .

The operations of triangular fuzzy numbers presented by Dubois and Prade [10] are based on the idea of three positions of each triangular fuzzy number $T = (a - b, a, a + c)$, where a means the middle point, b represents the length of interval to the membership 0 on the left side, and c denotes the length of interval to the membership 0 on the right side. Similarly, for two triangular fuzzy numbers $T_1 = (a_1 - b_1, a_1, a_1 + c_1)$ and $T_2 = (a_2 - b_2, a_2, a_2 + c_2)$, a_1 and a_2 represent the middle points of T_1 and T_2 , respectively, $a_1 - b_1$ and $a_2 - b_2$ represent the left endpoints of T_1 and T_2 , respectively, and $a_1 + c_1$ and $a_2 + c_2$ denote the right endpoints of T_1 and T_2 , respectively. Let λ be a positive constant number. The following operations hold:

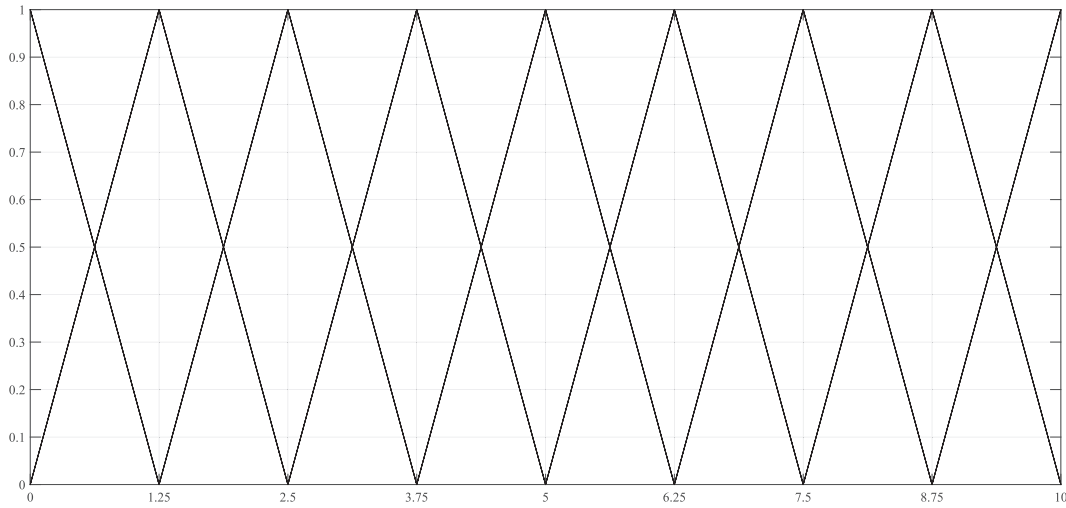


Fig. 2. Triangular membership functions of nine linguistic terms.

$$T_1 + T_2 = (a_1 - b_1 + a_2 - b_2, a_1 + a_2, a_1 + c_1 + a_2 + c_2)$$

$$T_1 - T_2 = (a_1 - b_1 - a_2 - c_2, a_1 - a_2, a_1 + c_1 - a_2 + b_2)$$

$$\lambda T_1 = (\lambda a_1, \lambda b_1, \lambda c_1)$$

$$-\lambda T_1 = (-\lambda a_1, -\lambda c_1, -\lambda b_1)$$

$$\min(T_1, T_2) = (\min(a_1 - b_1, a_2 - b_2), \min(a_1, a_2), \min(a_1 + c_1, a_2 + c_2))$$

$\min(T_1, T_2)$ is based on the pointwise minimum in the triangular fuzzy numbers. A fuzzy constraint is established by the inequality relations, $\tilde{\leq}$ and $\tilde{\geq}$, between fuzzy numbers [17,29], meaning “approximately less than or equal to” and “approximately greater than or equal to”, respectively. For two triangular fuzzy numbers $T_1 = (a_1 - b_1, a_1, a_1 + c_1)$ and $T_2 = (a_2 - b_2, a_2, a_2 + c_2)$, the inequality $T_1 \tilde{\leq} T_2$ can be modelled by $T_1^\alpha \leq T_2^\alpha$, where T_1^α and T_2^α are the alpha-cuts of the fuzzy numbers T_1 and T_2 , respectively, with α ranging from zero to one. Motivated by the idea of [17], the endpoints of fuzzy numbers with α being zero, i.e., $(a_1 - b_1)^\alpha = 0$, $(a_2 - b_2)^\alpha = 0$, $(a_1 + c_1)^\alpha = 0$, $(a_2 + c_2)^\alpha = 0$ and the most possible point with α being one, i.e., $a_1^\alpha = 1$, $a_2^\alpha = 1$, capture the characteristics of fuzzy numbers. $a_1 \leq a_2$, $a_1 - b_1 \leq a_2 - b_2$, and $a_1 + c_1 \leq a_2 + c_2$ are three conditions of the inequality relation between two fuzzy numbers based on the alpha-cut.

Based on the triangular fuzzy membership functions and the operations of triangular fuzzy numbers, we propose the probabilistic linguistic membership function with complete and incomplete probability information. The latter allows the ignorance in the probabilistic linguistic information, which means the sum of probabilities can be less than one.

The probabilistic linguistic information is complete without ignorance, such as $\tilde{v}$ is denoted as “30% good (s_6) and 70% very good (s_7)” with the nine granularity Γ in the uniform linguistic term set $\{s_0, s_1, \dots, s_7, s_8\}$ and the universe of discourse $[0, \delta]$, where Γ means the number of linguistic terms. The triangular fuzzy membership function of a linguistic term s_k is $s_k = \left(\frac{(k-1)\delta}{\Gamma-1}, \frac{k\delta}{\Gamma-1}, \frac{(k+1)\delta}{\Gamma-1} \right)$. The triangular fuzzy membership functions of “good” and “very good” are $s_6 = (5\delta/(\Gamma-1), 6\delta/(\Gamma-1), 7\delta/(\Gamma-1))$ and $s_7 = (6\delta/(\Gamma-1), 7\delta/(\Gamma-1), 8\delta/(\Gamma-1))$. Then, the function of the aggregated probabilistic linguistic pay off value can be obtained as $\tilde{v} = 0.3 * s_6 + 0.7 * s_7 = (5.7\delta/(\Gamma-1), 6.7\delta/(\Gamma-1), 7.7\delta/(\Gamma-1))$.

The unknown probability can be computed by $1 - \sum_{k=1}^K p_k$, where K is the number of linguistic terms in a probabilistic linguistic term set. The unknown probability can be assigned to the singleton linguistic term, the envelope of the probabilistic linguistic information such as $[s_6, s_7]$ in the above example, or the full linguistic term set [13]. In this paper, we suppose that the incomplete probabilistic linguistic information $\gamma = \{p_k | k = 1, 2, \dots, K\}$ can be normalized as $\gamma = \left\{ \left(p_k + \left(1 - \sum_{k=1}^K p_k \right) / K \right) s_k \mid k = 1, 2, \dots, K \right\}$. Thus, we have

$$\gamma = (a_\gamma - b_\gamma, a_\gamma, a_\gamma + c_\gamma) = \sum_{k=1}^K s_k \times \left[p_k + \left(1 - \sum_{k=1}^K p_k \right) / K \right] \quad (4)$$

For example, if a piece of probabilistic linguistic information “30% good (s_6) and 50% very good (s_7)” is given, then the unknown probability $1 - \sum_{k=1}^K p_k = 0.2$. As a result, the normalized probabilistic linguistic term can be obtained as $\{0.4 s_6, 0.6 s_7\}$. Using Eq. (4), the three parameters of the triangular membership function for denoting $\{0.4 s_6, 0.6 s_7\}$ are $(7, 8.25, 9.5)$, which can be visually shown in Fig. 3.

Based on the transformed triangular membership function of probabilistic linguistic information, the α -cut of the probabilistic linguistic information can be deduced and the value of α ranges from zero to one. If $\alpha = 0$, then the cut interval of the probabilistic linguistic information is $[a_{\tilde{v}} - b_{\tilde{v}}, a_{\tilde{v}} + c_{\tilde{v}}]$. If $\alpha = 1$, then the cut degrades to the crisp midpoint $a_{\tilde{v}}$. Actually, these three points $(a_{\tilde{v}} - b_{\tilde{v}}, 0)$, $(a_{\tilde{v}}, 1)$, and $(a_{\tilde{v}} + c_{\tilde{v}}, 0)$ determine the shape of the left and right membership functions with respect to the midpoint $a_{\tilde{v}}$.

3.3. Probabilistic linguistic two-person and zero-sum matrix game

In this subsection, fuzzy linear programming with probabilistic linguistic information is presented and fused into a constant-sum matrix game with probabilistic linguistic information.

3.3.1. For the gaining floor: Player I

Due to the partial ordering of fuzzy numbers, the objective in Eq. (2), $\max \{\tilde{v}_1\}$, is converted into $\max \{a_{\tilde{v}_1}, a_{\tilde{v}_1} - b_{\tilde{v}_1}, a_{\tilde{v}_1} + c_{\tilde{v}_1}\}$, which can be regarded as a multi-objective optimization problem. There are several methods to solve multi-objective linear optimization problems, such as the lexicographic optimization [5,17], evolutionary algorithms [12], and goal programming [24]. In this paper, the lexicographic optimization is used. The midpoint value $a_{\tilde{v}_1}$ is set as the most important objective, while the left two values $a_{\tilde{v}_1} - b_{\tilde{v}_1}$ and $a_{\tilde{v}_1} + c_{\tilde{v}_1}$ are regarded as the same second most important goals. Then, Eq. (2) can be transformed into Eq. (5) with probabilistic linguistic membership functions.

$$\begin{aligned} & \max a_{\tilde{v}_1} \\ \text{s.t. : } & \begin{cases} \sum_{i=1}^m (a_{\tilde{v}_{ij}} - b_{\tilde{v}_{ij}}) x_i \geq a_{\tilde{v}_1} - b_{\tilde{v}_1}, j = 1, 2, \dots, n \\ \sum_{i=1}^m a_{\tilde{v}_{ij}} x_i \geq a_{\tilde{v}_1}, j = 1, 2, \dots, n \\ \sum_{i=1}^m (a_{\tilde{v}_{ij}} + c_{\tilde{v}_{ij}}) x_i \geq a_{\tilde{v}_1} + c_{\tilde{v}_1}, j = 1, 2, \dots, n \\ \sum_{i=1}^m x_i = 1; x_i \geq 0, i = 1, 2, \dots, m \\ b_{\tilde{v}_1} \geq 0, c_{\tilde{v}_1} \geq 0 \end{cases} \end{aligned} \quad (5)$$

The optimized variable in Eq. (5) is $a_{\tilde{v}_1}$, the midpoint of $(a_{\tilde{v}_1} - b_{\tilde{v}_1}, a_{\tilde{v}_1}, a_{\tilde{v}_1} + c_{\tilde{v}_1})$. Solving Eq. (5), we can obtain the optimal value $a_{\tilde{v}_1}^*$, the preliminary left and right deviations, $b_{\tilde{v}_1}^0$ and $c_{\tilde{v}_1}^0$. Based on the results of Eqs. (5) and (6) is devised to find the maximal fuzzy gain value for the player I.

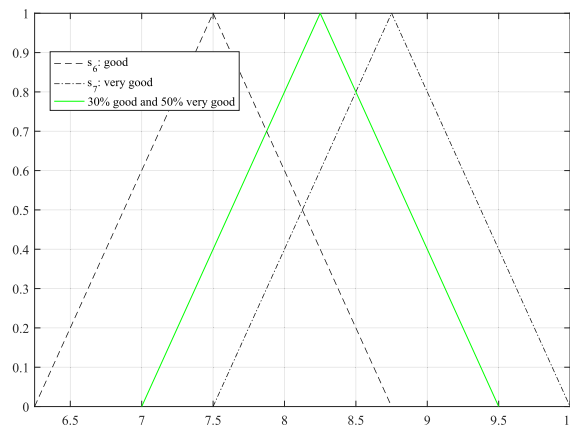


Fig. 3. The triangular membership function of the incomplete information.

$$\begin{aligned}
& \max 2a_{v_1}^* - b_{v_1} + c_{v_1} \\
& \text{s.t. : } \begin{cases} \sum_{i=1}^m (a_{v_{ij}} - b_{v_{ij}})x_i \geq a_{v_1}^* - b_{v_1}, j = 1, 2, \dots, n \\ \sum_{i=1}^m a_{v_{ij}} x_i \geq a_{v_1}^*, j = 1, 2, \dots, n \\ \sum_{i=1}^m (a_{v_{ij}} + c_{v_{ij}})x_i \geq a_{v_1}^* + c_{v_1}, j = 1, 2, \dots, n \\ 0 \leq b_{v_1} \leq b_{v_1}^0, 0 \leq c_{v_1} \leq c_{v_1}^0 \\ \sum_{i=1}^m x_i = 1; x_i \geq 0, i = 1, 2, \dots, m \end{cases} \quad (6)
\end{aligned}$$

The optimized variables in Eq. (6) are $-b_{v_1}$ and c_{v_1} because $a_{v_1}^* - b_{v_1}$ and $a_{v_1}^* + c_{v_1}$ are the left and right value with the membership degree, zero. The inequalities $0 \leq b_{v_1} \leq b_{v_1}^0$, $0 \leq c_{v_1} \leq c_{v_1}^0$ guarantee that the computed results of Eq. (6) satisfy the constraints in Eq. (5) because $a_{v_1}^*$, $b_{v_1}^0$ and $c_{v_1}^0$ are the calculated values in Eq. (5). Eq. (6) would produce the optimal strategy x^* and fuzzy value $\tilde{v}_1^* = (a_{v_1}^* - b_{v_1}^*, a_{v_1}^*, a_{v_1}^* + c_{v_1}^*)$ for the player I.

3.3.2. For the loss ceiling: The player II

Similarly, we set the midpoint value a_{v_2} as the most important objective. Then, Eq. (7) is a specific form of Eq. (3) with triangular membership functions for the probabilistic linguistic information:

$$\begin{aligned}
& \min a_{v_2} \\
& \text{s.t. : } \begin{cases} \sum_{j=1}^n (a_{v_{ij}} - b_{v_{ij}})y_j \leq a_{v_2} - b_{v_2}, i = 1, 2, \dots, m \\ \sum_{j=1}^n a_{v_{ij}} y_j \leq a_{v_2}, i = 1, 2, \dots, m \\ \sum_{j=1}^n (a_{v_{ij}} + c_{v_{ij}})y_j \leq a_{v_2} + c_{v_2}, i = 1, 2, \dots, m \\ \sum_{j=1}^n y_j = 1; y_j \geq 0, j = 1, 2, \dots, n \\ b_{v_2} \geq 0, c_{v_2} \geq 0 \end{cases} \quad (7)
\end{aligned}$$

The optimized variable in Eq. (7) is a_{v_2} , which is the value with the membership degree, one, in $(a_{v_2} - b_{v_2}, a_{v_2}, a_{v_2} + c_{v_2})$. Then, Eq. (7) is required to be solved before Eq. (8). The results of Eq. (7), the strategy y^* and the preliminary fuzzy value $(a_{v_2}^* - b_{v_2}^0, a_{v_2}^*, a_{v_2}^* + c_{v_2}^0)$, are the inputs of Eq. (8).

$$\begin{aligned}
& \min 2a_{v_2}^* - b_{v_2} + c_{v_2} \\
& \text{s.t. : } \begin{cases} \sum_{j=1}^n (a_{v_{ij}} - b_{v_{ij}})y_j \leq a_{v_2}^* - b_{v_2}, i = 1, 2, \dots, m \\ \sum_{j=1}^n a_{v_{ij}} y_j \leq a_{v_2}^*, i = 1, 2, \dots, m \\ \sum_{j=1}^n (a_{v_{ij}} + c_{v_{ij}})y_j \leq a_{v_2}^* + c_{v_2}, i = 1, 2, \dots, m \\ 0 \leq b_{v_2} \leq b_{v_2}^0, 0 \leq c_{v_2} \leq c_{v_2}^0 \\ \sum_{j=1}^n y_j = 1; y_j \geq 0, j = 1, 2, \dots, n \end{cases} \quad (8)
\end{aligned}$$

The optimized variables in Eq. (7) are $-b_{v_2}$ and c_{v_2} , which considers the left value $a_{v_2}^* - b_{v_2}$ and the right value $a_{v_2}^* + c_{v_2}$ as the same importance by adding $a_{v_2}^* - b_{v_2}$ and $a_{v_2}^* + c_{v_2}$ together in the objective. The minimal objective $\min 2a_{v_2}^* - b_{v_2} + c_{v_2}$ and the inequalities $0 \leq b_{v_2} \leq b_{v_2}^0$, $0 \leq c_{v_2} \leq c_{v_2}^0$ are devised to the solution of Eq. (8) in the solutions space of Eq. (7). Eq. (8) outputs the optimal strategy y^* and the corresponding optimal fuzzy value $\tilde{v}_2^* = (a_{v_2}^* - b_{v_2}^*, a_{v_2}^*, a_{v_2}^* + c_{v_2}^*)$ for the player II.

3.3.3. The solution of the probabilistic linguistic matrix game

The solution of the probabilistic linguistic matrix game is defined as the minimal value of v_1^* and v_2^* based on the point-wise minimum as follows:

$$\begin{aligned}
v^* &= (a_{v^*}^* - b_{v^*}^*, a_{v^*}^*, a_{v^*}^* + c_{v^*}^*) = \min(v_1^*, v_2^*) \\
&= \left(\min(a_{v^*}^* - b_{v^*}^*, a_{v^*}^* - b_{v^*}^*), \min(a_{v^*}^*, a_{v^*}^*), \min(a_{v^*}^* + c_{v^*}^*, a_{v^*}^* + c_{v^*}^*) \right) \quad (9)
\end{aligned}$$

The optimal fuzzy value v^* of the probabilistic linguistic game where the probabilistic linguistic information is modelled as a triangular membership function. The strategies x^* and y^* related to the optimal values of v_1^* and v_2^* are called as the maximin and minimax strategy for the player I and the player II, respectively. The strategy portfolio (x^*, y^*) is referred as the saddle point as well.

After representing probabilistic linguistic information by triangular fuzzy membership functions and obtaining the corresponding optimal fuzzy values as above, we can then retranslate the optimal fuzzy values into probabilistic linguistic information. The retranslation process contains two steps:

- 1) Find the Top 2 nearest linguistic terms, $\underline{s}_{a_v^*}$ and $\bar{s}_{a_v^*}$, around the optimal midpoint value a_v^* . In this step, we need to calculate the distance between the optimal midpoint value a_v^* and the semantics of linguistic terms $I(s_k)$. The Top 2 minimal elements in the set $\{|a_v^* - k\delta/(\Gamma - 1)| | k = 0, 1, 2, \dots, \Gamma - 1\}$ show the nearest distance around the optimal value a_v^* . By the ordered relation of the upper bound $\bar{s}_{a_v^*}$ and the lower bound $\underline{s}_{a_v^*}$, $\underline{a}_v^* < \bar{a}_v^*$, two adjacent linguistic terms $\underline{s}_{a_v^*}$ and $\bar{s}_{a_v^*}$ can be obtained.
- 2) Based on the linear interpolation method, $a_v^* = \theta \bar{a}_v^* + (1 - \theta) \underline{a}_v^*$, we can find the proportion θ to combine the two adjacent linguistic terms:

$$\theta = \frac{\bar{a}_v^* - a_v^*}{\bar{a}_v^* - \underline{a}_v^*}$$

The fuzzy game value can be transformed into the probabilistic linguistic information, $\{\theta \underline{s}_{a_v^*}, (1 - \theta) \bar{s}_{a_v^*}\}$, which is more understandable than the traditional fuzzy game. The pseudocode mode of the retranslation process is organized in Fig. 4.

In summary, this section represents the membership function of probabilistic linguistic information by triangular fuzzy membership functions. Then, the defuzzied probabilistic linguistic information is introduced into the matrix game for allowing uncertain linguistic evaluations in the payoff matrix. After that, the saddle point, consisting of the optimal strategy for the player I and the player II, can be found with the corresponding fuzzy game values by the proposed optimization models. Finally, the fuzzy game values are retranslated into the understandable probabilistic linguistic information. The retranslation process of probabilistic linguistic information is useful for computing with words as well.

4. An illustrative example: Forest management

In this section, an example about the forest management is given to illustrate the usefulness and applicability of the two-person matrix game with probabilistic linguistic information. This special example contains two competitive players: one is the decision-maker and another one is the Nature. The game with nature originates from the portfolio choice problem [31]. In this example, the forest management department is regarded as the player I and Nature is regarded as the player II to form the two-person matrix game.

Input	The linguistic term set $\{s_0, s_1, \dots, s_7, s_{\Gamma-1}\}$ and $v^* = (a_v^* - b_v^*, a_v^*, a_v^* + c_v^*)$
Step 1	For each linguistic term s_k in the linguistic term set Calculate the absolute distance between a_v^* and s_k by $a_v^* - k\delta/(\Gamma - 1)$
Step 2	Find top 2 minimal elements in the distance set $\{ a_v^* - k\delta/(\Gamma - 1) k = 0, 1, 2, \dots, \Gamma - 1\}$ Record indices of top 2 minimal elements, $\underline{a}_v^*$ and $\bar{a}_v^*$
Step 3	Compute the proportion θ by $\theta = \frac{\bar{a}_v^* - a_v^*}{\bar{a}_v^* - \underline{a}_v^*}$
Output	The probabilistic linguistic game value $\{\theta \underline{s}_{a_v^*}, (1 - \theta) \bar{s}_{a_v^*}\}$

Fig. 4. The pseudocode mode of the retranslation process.

4.1. Case description

In the forest management, different objectives may conflict with each other [11,25]. The forest management department may pay more attention to economic returns than other objectives, such as the carbon storage, forest biodiversity and habitat availability. However, from the perspective of the player II, Nature is against with the player I on economic returns. Business as usual, set aside and continuous cover forestry are three management regimes devised for the forest management department. The three different management regimes can be viewed as the three strategies of the player I in the game. In a similar manner, carbon storage, forest biodiversity, economic returns and habitat availability are four strategies of the player II, Nature. The payoff matrix values with probabilistic linguistic information are obtained in Table 1.

The evaluation information in Table 1 can be depicted in the forms of probabilistic linguistic term sets in the payoff matrix $\tilde{\mathbf{V}}$.

$$\tilde{\mathbf{V}} = \begin{pmatrix} \{s_6(0.6), s_7(0.4)\} & \{s_5(1)\} & \{s_6(0.2), s_7(0.8)\} & \{s_4(0.3), s_5(0.5)\} \\ \{s_5(0.1), s_6(0.9)\} & \{s_5(0.8), s_6(0.2)\} & \{s_5(0.7), s_6(0.3)\} & \{s_5(0.5), s_6(0.5)\} \\ \{s_4(0.5), s_5(0.5)\} & \{s_6(0.5), s_7(0.5)\} & \{s_6(1)\} & \{s_6(0.3), s_7(0.7)\} \end{pmatrix}$$

In the following, we will solve this multiple criteria analysis problem with the probabilistic linguistic matrix game and then compare it with other methods.

4.2. Solving the case by the probabilistic linguistic matrix game

After obtaining the probabilistic linguistic payoff value matrix, we can defuzzify the probabilistic linguistic information by membership functions. Based on the presented triangular membership functions of the probabilistic linguistic information in Section 3.2, the three parameters of each triangular membership function corresponding to each probabilistic linguistic term can be computed and shown in Table 2.

Then, the objective models, Eqs. (11) and (12), are established based on the probabilistic linguistic linear programming models proposed in Section 3.3 for the player I and the player II, respectively. The solutions of Eqs. (11) and (12) are the preliminary inputs of Eqs. (13) and (14) for the player I and the player II, respectively, which are detailed as follows:

$$\begin{aligned} & \max a_{\tilde{v}_1} \\ \text{s.t. : } & \begin{cases} 6.75x_1 + 6.125x_2 + 4.375x_3 \geq a_{\tilde{v}_1} - b_{\tilde{v}_1}^0 \\ 5x_1 + 5.25x_2 + 6.875x_3 \geq a_{\tilde{v}_1} - b_{\tilde{v}_1}^0 \\ 7.25x_1 + 5.375x_2 + 6.25x_3 \geq a_{\tilde{v}_1} - b_{\tilde{v}_1}^0 \\ 4.5x_1 + 5.625x_2 + 7.125x_3 \geq a_{\tilde{v}_1} - b_{\tilde{v}_1}^0 \\ 8x_1 + 7.375x_2 + 5.625x_3 \geq a_{\tilde{v}_1} \\ 6.25x_1 + 6.5x_2 + 8.125x_3 \geq a_{\tilde{v}_1} \\ 8.5x_1 + 6.625x_2 + 7.5x_3 \geq a_{\tilde{v}_1} \\ 5.75x_1 + 6.875x_2 + 8.375x_3 \geq a_{\tilde{v}_1} \\ 9.25x_1 + 8.625x_2 + 6.875x_3 \geq a_{\tilde{v}_1} + c_{\tilde{v}_1}^0 \\ 7.5x_1 + 7.75x_2 + 9.375x_3 \geq a_{\tilde{v}_1} + c_{\tilde{v}_1}^0 \\ 9.75x_1 + 7.875x_2 + 8.75x_3 \geq a_{\tilde{v}_1} + c_{\tilde{v}_1}^0 \\ 7x_1 + 8.125x_2 + 9.625x_3 \geq a_{\tilde{v}_1} + c_{\tilde{v}_1}^0 \\ \sum_{i=1}^3 x_i = 1; x_i \geq 0, i = 1, 2, 3 \\ b_{\tilde{v}_1}^0 \geq 0, c_{\tilde{v}_1}^0 \geq 0 \end{cases} \end{aligned} \quad (11)$$

Table 1

The probabilistic linguistic payoff matrix values of three different management regimes.

Payoff values	Carbon storage	Forest biodiversity	Economic returns	Habitat availability
Business as usual	60% G and 40% VG	SG	20% G and 80% VG	30% M and 50% SG
Set aside	10% SG and 90% G	80% SG and 20% G	70% SG and 30% G	Between SG and G
Continuous cover forestry	Between M and SG	Between G and VG	G	30% G and 70% VG

Note: VG (Very Good), G(Good), SG (Slightly Good), M(Medium).

Table 2

Triangular payoff matrix values based on the probabilistic linguistic membership function.

Payoff values	Carbon storage	Forest biodiversity	Economic returns	Habitat availability
Business as usual	(6.75,8,9.25)	(5.6,25,7.5)	(7.25,8.5,9.75)	(4.5,5.75,7)
Set aside	(6.125,7.375,8.625)	(5.375,6.625,7.875)	(5.375,6.625,7.875)	(5.625,6.875,8.125)
Continuous cover forestry	(4.375,5.625,6.875)	(6.875,8.125,9.375)	(6.25,7.5,8.75)	(7.125,8.375,9.625)

Solving Eq. (11) for the player I, we can obtain the initial results: $x_1^0 = 0.3776$, $x_2^0 = 0.2653$, $x_3^0 = 0.3571$, $a_{\tilde{v}_1} = 6.9860$, $b_{\tilde{v}_1}^0 = 1.2500$, and $c_{\tilde{v}_1}^0 = 0.0000$.

$$\begin{aligned}
 & \min a_{\tilde{v}_2} \\
 \text{s.t. : } & \left\{ \begin{array}{l} 6.75y_1 + 5y_2 + 7.25y_3 + 4.5y_4 \leq a_{\tilde{v}_2} - b_{\tilde{v}_2}^0 \\ 6.125y_1 + 5.25y_2 + 5.375y_3 + 5.625y_4 \leq a_{\tilde{v}_2} - b_{\tilde{v}_2}^0 \\ 4.375y_1 + 6.875y_2 + 6.25y_3 + 7.125y_4 \leq a_{\tilde{v}_2} - b_{\tilde{v}_2}^0 \\ 8y_1 + 6.25y_2 + 8.5y_3 + 5.75y_4 \leq a_{\tilde{v}_2} \\ 7.375y_1 + 6.5y_2 + 6.625y_3 + 6.875y_4 \leq a_{\tilde{v}_2} \\ 5.625y_1 + 8.125y_2 + 7.5y_3 + 8.375y_4 \leq a_{\tilde{v}_2} \\ 9.25y_1 + 7.5y_2 + 9.75y_3 + 7y_4 \leq a_{\tilde{v}_2} + c_{\tilde{v}_2}^0 \\ 8.625y_1 + 7.75y_2 + 7.875y_3 + 8.125y_4 \leq a_{\tilde{v}_2} + c_{\tilde{v}_2}^0 \\ 6.875y_1 + 9.375y_2 + 8.75y_3 + 9.625y_4 \leq a_{\tilde{v}_2} + c_{\tilde{v}_2}^0 \\ \sum_{j=1}^4 y_j = 1; y_j \geq 0, j = 1, 2, 3, 4 \\ b_{\tilde{v}_2} \geq 0, c_{\tilde{v}_2} \geq 0 \end{array} \right. \quad (12)
 \end{aligned}$$

Similarly, $y_1^0 = 0.4745$, $y_2^0 = 0.3367$, $y_3^0 = 0.0000$, $y_4^0 = 0.1888$, $a_{\tilde{v}_2} = 6.9860$, $b_{\tilde{v}_2}^0 = 0.0000$, $c_{\tilde{v}_2}^0 = 1.2500$ can be obtained by Eq. (12) for the player II.

Then, the deviations of the players' game values are optimized by Eqs. (13) and (14) on the basis of the results of Eqs. (11) and (12), shown as follows:

$$\begin{aligned}
 & \max 2a_{\tilde{v}_1}^* - b_{\tilde{v}_1} + c_{\tilde{v}_1} \\
 \text{s.t. : } & \left\{ \begin{array}{l} 6.75x_1 + 6.125x_2 + 4.375x_3 \geq 6.9860 - b_{\tilde{v}_1} \\ 5x_1 + 5.25x_2 + 6.875x_3 \geq 6.9860 - b_{\tilde{v}_1} \\ 7.25x_1 + 5.375x_2 + 6.25x_3 \geq 6.9860 - b_{\tilde{v}_1} \\ 4.5x_1 + 5.625x_2 + 7.125x_3 \geq 6.9860 - b_{\tilde{v}_1} \\ 8x_1 + 7.375x_2 + 5.625x_3 \geq 6.9860 \\ 6.25x_1 + 6.5x_2 + 8.125x_3 \geq 6.9860 \\ 8.5x_1 + 6.625x_2 + 7.5x_3 \geq 6.9860 \\ 5.75x_1 + 6.875x_2 + 8.375x_3 \geq 6.9860 \\ 9.25x_1 + 8.625x_2 + 6.875x_3 \geq 6.9860 + c_{\tilde{v}_1} \\ 7.5x_1 + 7.75x_2 + 9.375x_3 \geq 6.9860 + c_{\tilde{v}_1} \\ 9.75x_1 + 7.875x_2 + 8.75x_3 \geq 6.9860 + c_{\tilde{v}_1} \\ 7x_1 + 8.125x_2 + 9.625x_3 \geq 6.9860 + c_{\tilde{v}_1} \\ 0 \leq b_{\tilde{v}_1} \leq 1.25; c_{\tilde{v}_1} \geq 0 \\ \sum_{i=1}^3 x_i = 1; x_i \geq 0, i = 1, 2, 3 \end{array} \right. \quad (13)
 \end{aligned}$$

The optimal strategy for the player I is $x_1^* = 0.3776$, $x_2^* = 0.2653$, and $x_3^* = 0.3571$. The corresponding game value $\tilde{v}_1$ of the player I is $a_{\tilde{v}_1}^* = 6.9860$, $b_{\tilde{v}_1}^* = 0.0000$, and $c_{\tilde{v}_1}^* = 15.2220$.

$$\begin{aligned}
 & \min 2a_{\tilde{v}_2}^* - b_{\tilde{v}_2}^* + c_{\tilde{v}_2}^* \\
 \text{s.t. : } & \begin{cases} 6.75y_1 + 5y_2 + 7.25y_3 + 4.5y_4 \leq 6.9860 - b_{\tilde{v}_2} \\ 6.125y_1 + 5.25y_2 + 5.375y_3 + 5.625y_4 \leq 6.9860 - b_{\tilde{v}_2} \\ 4.375y_1 + 6.875y_2 + 6.25y_3 + 7.125y_4 \leq 6.9860 - b_{\tilde{v}_2} \\ 8y_1 + 6.25y_2 + 8.5y_3 + 5.75y_4 \leq 6.9860 \\ 7.375y_1 + 6.5y_2 + 6.625y_3 + 6.875y_4 \leq 6.9860 \\ 5.625y_1 + 8.125y_2 + 7.5y_3 + 8.375y_4 \leq 6.9860 \\ 9.25y_1 + 7.5y_2 + 9.75y_3 + 7y_4 \leq 6.9860 + c_{\tilde{v}_2} \\ 8.625y_1 + 7.75y_2 + 7.875y_3 + 8.125y_4 \leq 6.9860 + c_{\tilde{v}_2} \\ 6.875y_1 + 9.375y_2 + 8.75y_3 + 9.625y_4 \leq 6.9860 + c_{\tilde{v}_2} \\ \sum_{j=1}^4 y_j = 1; y_j \geq 0, j = 1, 2, 3, 4 \\ b_{\tilde{v}_2} \geq 0, 1.25 \geq c_{\tilde{v}_2} \geq 0 \end{cases} \quad (14)
 \end{aligned}$$

Similarly, the player II's optimal strategy and game values are found as $y_1^* = 0.4745$, $y_2^* = 0.3367$, $y_3^* = 0.0000$, $y_4^* = 0.1888$, $\tilde{v}_2 = (a_{\tilde{v}_2}^*, b_{\tilde{v}_2}^*, c_{\tilde{v}_2}^*) = (5.736, 6.9860, 8.236)$. That is to say, Nature thinks the economic returns are not important at all and views the carbon dioxide, forest biodiversity and habitat availability as three main factors. The applications of the game theory in multiple criteria analysis does not require the importance values of the factors and outputs the latent priorities of criteria.

The optimal game value v^* can be obtained as the intersection of players' game values, v_1^* and v_2^* , shown as follows:

$$\mu(v^*) = \min\{\mu(v_1^*), \mu(v_2^*)\} = (5.736, 6.9860, 8.236)$$

Then, the game value v^* can be retranslated by Eq. (10) into the probabilistic linguistic term $\{0.4112 s_5, 0.5888 s_6\}$ and the probabilistic linguist evaluation information as “41.12% slightly good and 58.88% good”. The recommended strategy of the forest management department is (0.3776, 0.2653, 0.3571) rather than the pure strategy which only adopts each management at one time.

4.3. Comparative analyses and discussions

In this subsection, the comparative analyses with the previous games with 2-tuple linguistic information and the probabilistic linguistic multiple criteria analysis are given. Some managerial implications are provided for the forest management.

4.3.1. Compared with the 2-tuple linguistic game

The comparisons are implemented from the information representation and the application in multiple criteria analysis.

Compared with the 2-tuple linguistic information, the probabilistic linguistic information allows players to provide evaluations on several linguistic terms rather than only one linguistic term. Besides, the 2-tuple linguistic information can not accept or handle incomplete evaluations, which would increase the burden and difficulty for players. Moreover, the probabilistic linguistic information is defuzzified into simpler triangular membership functions instead of the linguistic symbols in the 2-tuple linguistic information. The same input data in Table 1 is described as $\tilde{V}_{2p}$ in the 2-tuple linguistic information with one linguistic term, where the second number means the deviation to the first linguistic term. The right column item is $(s_4, -0.3)$ because of the inability to handle the incomplete probability information.

$$\tilde{V}_{2p} = \begin{pmatrix} (s_6, 0.4) & (s_5, 0) & (s_7, -0.2) & (s_4, -0.3) \\ (s_6, -0.1) & (s_5, 0.2) & (s_5, 0.1) & (s_5, 0.5) \\ (s_4, 0.5) & (s_6, 0.5) & (s_6, 0) & (s_7, -0.3) \end{pmatrix}$$

Using the model in the game with 2-tuple linguistic information, Eq. (15) can be obtained as follows:

$$\begin{aligned}
 & \min V_{2p}^+ = X_1 + X_2 + X_3 \\
 \text{s.t. : } & \begin{cases} 6.4X_1 + 5.9X_2 + 4.5X_3 \geq 4/5 \\ 5X_1 + 5.2X_2 + 6.5X_3 \geq 4/5 \\ 6.8X_1 + 5.1X_2 + 6X_3 \geq 4/5 \\ 3.7X_1 + 5.5X_2 + 6.7X_3 \geq 4/5 \\ X_1, X_2, X_3 \geq 0 \end{cases}, \quad \text{s.t. : } \begin{cases} \max V_{2p}^- = Y_1 + Y_2 + Y_3 + Y_4 \\ 6.4Y_1 + 5Y_2 + 6.8Y_3 + 3.7Y_4 \leq 4/5 \\ 5.9Y_1 + 5.2Y_2 + 5.1Y_3 + 5.5Y_4 \leq 4/5 \\ 4.5Y_1 + 6.5Y_2 + 6Y_3 + 6.7Y_4 \leq 4/5 \\ Y_1, Y_2, Y_3, Y_4 \geq 0 \end{cases} \quad (15)
 \end{aligned}$$

After solving Eq. (15) by the method in [33], the optimal strategies are (0.1083, 0.6004, 0.2913) and (0.4258, 0.0000, 0.3100, 0.2642) for the players I and II, respectively. Singh and Gupta [33] viewed the player I's optimal strategy as the priority of three alternatives and output the ranking of alternatives based on the priority values. That is to say, Set aside A_2 > Continuous cover forestry A_3 > Business as usual A_1 .

On the one hand, if we replace the value $(s_4, -0.3)$ with $(s_5, -0.4)$ which is calculated by Eq. (4) to assign the unknown probability information in the payoff value, the optimal strategy can be obtained as (0.3776, 0.2653, 0.3571) and (0.4745, 0.3367, 0.0000, 0.1888) for the players I and II, respectively. In other words, [33] recommends the ranking, Business as usual A_1 > Continuous cover forestry A_3 > Set aside A_2 . The latter strategy with $(s_5, -0.4)$ is concordant with the result computed in Section 4.2, which shows the importance of capturing all information carefully.

On the other hand, the recommended ranking of alternatives may ignore the original definition of mixed strategies for a game. In our case study, the obtained strategies are still treated as a whole. The recommended strategy for the forest management department is a mixed combination of three strategies.

4.3.2. Compared with multiple criteria analysis methods

The comparisons with multiple criteria analysis methods are on the basis of the input and output. In multiple criteria analysis, players are asked to use the prior expertise information to provide the weights of criteria, whereas the main characteristic of the two-person matrix game with probabilistic linguistic information is that, it does not need the input of weights. The probabilistic linguistic matrix game introduces Nature as the competitive player II to tackle this issue. From the point of view of a multi-criteria issue, the case study would be modeled in terms of criteria as carbon storage, forest biodiversity, economic returns and habitat availability. Alternatives are business as usual, set aside, and continuous cover forestry. The criteria are assumed as equal importance with 0.25. The scores of three alternatives are (5.875, 7.125, 8.375), (5.625, 6.785, 8.125) and (6.156, 7.406, 8.656). The ranking of alternatives is Continuous cover forestry A_3 > Business as usual A_1 > Set aside A_2 . In addition, multiple criteria analysis usually outputs the ranking of alternatives and recommends the first position to the player. The probabilistic linguistic matrix game recommends a combination (i.e., mix) of these alternatives (if possible) at first. If a specific ranking of alternatives is required to output, the priorities of alternatives are inferred from the aspect of the game with Nature rather than the aggregated performances with predetermined weights of criteria.

4.3.3. Managerial implications

In this example, the forest management is recommended to adopt the fixed-percentage combination (i.e., mixed) strategy instead of any single (i.e., pure) strategy. Considering that forest is not a very small land with trees, each individual strategy among the strategy space formed with business as usual, set aside and continuous forest cover, may be not suitable for the whole forest management. Using the data in Table 1, the recommendation for the forest management department is 37.76% business as usual, 26.53% set aside and 35.71% continuous forest cover. If the player wants to change the evaluation data in Table 1, the recommended strategy would vary to some extent.

5. Conclusions

This paper proposed and investigated the two-person matrix game with probabilistic linguistic information, which allows the game theory to accept the incomplete linguistic information as input. The triangular membership functions for probabilistic linguistic information were presented to defuzzify the probabilistic linguistic information. Then, linear programming models for two players with the defuzzified probabilistic linguistic information were proposed to obtain the optimal game value and corresponding mixed strategies of players. The calculated game values in the linear models were retranslated by the linear interpolation method into the probabilistic linguistic information. Afterwards, the validity and advantages of the probabilistic linguistic matrix game were verified and justified by applying it to solve a multiple criteria problem about the forest management from the perspective of the game with Nature.

As for future research directions, other types of probabilistic linguistic membership functions, such as Gaussian membership function, for defuzzification and the retranslation techniques are interesting research topics with challenges. Moreover, the evolutionary game with the natural language information [9,14] is a fascinating research idea as well.

CRediT authorship contribution statement

Xiaomei Mi: Conceptualization, Methodology, Formal analysis, Writing - original draft, Writing - review & editing. **Huchang Liao:** Conceptualization, Investigation, Writing - original draft, Writing - review & editing, Funding acquisition. **Xiao-Jun Zeng:** Writing - review & editing, Project administration. **Zeshui Xu:** Writing - review & editing, Project administration.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The work was supported by the National Natural Science Foundation of China (Nos. 71771156, 71971145), the Scholarship from Chicccna Scholarship Council (No. 201906240161), and the Fundamental Research Funds for the Central Universities (No. SCU-BS-PY-202007).

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